

Aryo Pradipta Gema

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SUMMARY

Aryo Gema is a PhD student in Biomedical AI at the University of Edinburgh, supervised by Dr. Beatrice Alex, Dr. Pasquale Minervini, and Dr. Luke Daines. His research lies at the intersection of AI safety and AI in healthcare, with a focus on understanding and mitigating what can go wrong in large language models (LLMs)—including hallucinations, reasoning failure modes, and potential misuse. His work aims to improve the reliability and trustworthiness of LLMs in safety-critical settings.

EDUCATION

University of Edinburgh

Edinburgh, United Kingdom

*PhD in Biomedical AI; **expected: August 2026***

Sep 2023 – Present

- **Thesis:** Domain adaptation, Hallucination mitigation, and Reasoning robustness of Clinical LLMs.
- **Advisors:** Dr. Beatrice Alex, Dr. Pasquale Minervini, and Dr. Luke Daines.

*MSc by Research in Biomedical AI; **Distinction***

Sep 2022 – Aug 2023

- **Thesis:** Parameter-efficient Fine-tuning of LLMs for the Clinical Domain.
- **Advisors:** Dr. Beatrice Alex, Dr. Pasquale Minervini, and Dr. Luke Daines.

Bina Nusantara University

Jakarta, Indonesia

*BSc in Computer Science; **GPA: 3.87/4.00***

Sep 2014 – Aug 2018

- **Thesis:** Sequence-To-Sequence Learning For Motion-Aware Claim Generation.
- **Advisors:** Dr. Derwin Suhartono.

PROFESSIONAL EXPERIENCE

Anthropic (AI safety and research company)

London, United Kingdom

AI Safety Fellow

March 2025 – November 2025

- Researched the failure modes of reasoning models (Inverse Scaling in Test-Time Compute).
- Researched the science of LLM misuse uplift.

AstraZeneca (Pharmaceutical and biotechnology company)

Cambridge, United Kingdom

Research Scientist Intern

July 2024 – September 2024

- Researched LLM hallucination mitigation, focusing on innovative post-training strategies.
- Developed and tested a novel decoding framework, DeCoRe, which leverages contrasting retrieval heads to reduce hallucinations in LLM outputs.

Epigene Labs (AI-driven Precision Oncology startup)

Paris, France

Senior Data Scientist

October 2021 – August 2022

- Supervised data collection, data labelling, ontology formulations, and model evaluations.
- Maintained the quality of more than 20 Machine Learning models in production.
- Worked with the Head of Data to align Data Science research and product timelines.
- Supervised data science internship projects on NLP-based biomedical concept extraction.
- Initiated a regular inter-departmental research workshop.

Data Scientist

October 2019 – September 2021

- Researched and implemented a Language-Agnostic, Format-Agnostic, and Cancer-Agnostic Data Harmonisation Pipeline using hybrid Machine Learning techniques.
- Supervised data science internship projects on Tailored Performance Metrics and Continuous Learning.
- Contribute to an open-sourced batch effect correction library (<https://github.com/epigenelabs/pyComBat>).
- Directly worked under the C-level to align data science research, DevOps, and product timelines.

Riminder (now HrFlow; Recruitment solution startup)

Paris, France

Staff AI Research Scientist

February 2019 – October 2019

- Supervised day-to-day operations of the research team.
- Directly worked with the C-level to align research and product timelines.

- Improved in-house Named Entity Recognition model by 20% F1 score, block tagger, and other ML models by 5%.

AI Research Scientist

October 2018 – January 2019

- Contributed to the research team to further improve profile-to-job matching/recommendation using deep learning algorithms.
- Introduced novel approaches for profile-to-job matching, and improved the model by a 10% ROC AUC score.
- Held ownership of multiple of Riminder's highest-valued projects.

Julo (AI-driven financial technology startup)

Jakarta, Indonesia

Data Scientist

September 2017 – July 2018

- Developed multiple machine learning solutions, including Automatic individual credit performance grading, deep learning-based supervised and unsupervised text processing, and recognition.
- Worked collaboratively with the Head of Analytics in day-to-day development and recruitment.
- Designed and supervised data collection, data labelling, and evaluations.
- Mentored junior data scientists and interns.

Bina Nusantara University

Jakarta, Indonesia

Undergraduate Researcher; placement programme

March 2017 – July 2018

- Researched cutting-edge Deep Learning and other Machine Learning techniques in Argumentation Mining.
- Published 3 papers in 2 international conferences and 1 journal (ICCCSCI 2017, MIWAI-SoDeLIA 2017, IJCVR 2018).
- Best Presenter of ICCSCI 2017.
- Appointed as the lead researcher for the team of undergraduate researchers.

RE/MAX 2000 (real-estate company)

California, United States (Remote)

Freelance Full Stack Software Engineer

July 2017 – September 2017

- Developed 10 core modules for internal Customer Relationship Management (CRM) application.
- Developed 8 core modules for Image and Descriptive Text Verification applications.
- Developed 5 WordPress-based websites.

Volantech (B2B software house)

Jakarta, Indonesia

Software Engineer

February 2016 – July 2017

- Developed Customer Relationship Management (CRM) applications for mining companies.
- Held ownership of multiple of Volantech's high-value projects.
- Worked closely with the C-level in developing product timelines.
- Research on Deep Learning techniques in conjunction with Internet of Things systems.

PUBLICATIONS

In proceedings

- **Aryo Pradipta Gema**, ..., Yanda Chen, Joe Benton, Ethan Perez (2025). *Inverse scaling in test-time compute*. **TMLR**.
- Neel Rajani, Aryo Pradipta Gema, Seraphina Goldfarb-Tarrant, Ivan Titov (2025). *Scalpel vs. Hammer: GRPO Amplifies Existing Capabilities, SFT Replaces Them*. **Actionable Interpretability @ ICML 2025**
- Jingjie Zheng, **Aryo Pradipta Gema**, Giwon Hong, Xuanli He, Pasquale Minervini, Youcheng Sun, Qionghai Xu (2025). *GRADA: Graph-based Reranker against Adversarial Documents Attack*. **EMNLP 2025**
- Mohammad Reza Ghasemi Madani, **Aryo Pradipta Gema**, Gabriele Sarti, Yu Zhao, Pasquale Minervini, Andrea Passerini (2025). *Noiser: Bounded Input Perturbations for Attributing Large Language Models*. **COLM 2025**
- Rohan Gorantla, **Aryo Pradipta Gema**, ..., Benjamin Suutari, Jordi Juárez Jiménez, Antonia SJS Mey (2024). *Learning Binding Affinities via Fine-tuning of Protein and Ligand Language Models*. **Journal of Chemical Information and Modeling**
- Ahmed Abdulaal, Chen Jin, Nina Montaña-Brown, **Aryo Pradipta Gema**, ..., Amrutha Saseendran (2025). *Balancing Act: Diversity and Consistency in Large Language Model Ensembles*. **ICLR 2025**
- Rohit Saxena, **Aryo Pradipta Gema**, Pasquale Minervini (2025). *Lost in Time: Clock and Calendar Understanding Challenges in Multimodal LLMs*. **Reasoning and Planning for Large Language Models @ ICLR 2025**
- Andrea Passerini, **Aryo Pradipta Gema**, Pasquale Minervini, Burcu Sayin, Katya Tentori (2025). *Fostering effective hybrid human-LLM reasoning and decision making*. **Frontiers in Artificial Intelligence**

- **Aryo Pradipta Gema**, Chen Jin, ..., Pasquale Minervini, Amrutha Saseendran (2024). *DeCoRe: Decoding by Contrasting Retrieval Heads to Mitigate Hallucinations*. **EMNLP 2025**
- Joshua Ong Jun Leang, **Aryo Pradipta Gema**, Shay B Cohen (2024). *CoMAT: Chain of mathematically annotated thought improves mathematical reasoning*. **EMNLP 2025**
- Ne Luo, **Aryo Pradipta Gema**, ..., Pasquale Minervini (2025). *Self-Training Large Language Models for Tool-Use Without Demonstrations*. **NAACL 2025**
- Yu Zhao, Alessio Devoto, Giwon Hong, Xiaotang Du, **Aryo Pradipta Gema**, ..., Kam-Fai Wong, Pasquale Minervini (2024). *Steering knowledge selection behaviours in llms via sae-based representation engineering*. **NAACL 2025**
- **Aryo Pradipta Gema**, Leang, J. O. J., Giwon Hong, Alessio Devoto, Mancino, A. C. M., Saxena, R., ..., Pasquale Minervini (2024). *Are We Done with MMLU?*. **NAACL 2025**
- Yu Zhao, Xiaotang Du, Giwon Hong, **Aryo Pradipta Gema**, ..., Kam-Fai Wong, Pasquale Minervini (2024). *Analysing the Residual Stream of Language Models Under Knowledge Conflicts*. arXiv preprint arXiv:2410.16090. **The Foundation Model Intervention Workshop @ NeurIPS 2024**
- Matus Falis, **Aryo Pradipta Gema**, Hang Dong, ..., Beatrice Alex. (2024). *Can GPT-3.5 generate and code discharge summaries?*. **JAMIA**
- **Aryo Pradipta Gema**, Dominik Grabarczyk, Wolf De Wulf, Piyush Borole, Javier Alfaro, Pasquale Minervini, Antonio Vergari, Ajitha Rajan (2024). *Knowledge Graph Embeddings in the Biomedical Domain: Are They Useful? A Look at Link Prediction, Rule Learning, and Downstream Polypharmacy Tasks*. **Bioinformatics Advances**
- **Aryo Pradipta Gema**, Chaeun Lee, Pasquale Minervini, Luke Daines, Ian Simpson, Beatrice Alex. (2024). Edinburgh clinical NLP at MEDIQA-CORR 2024: Guiding large language models with hints. **ClinicalNLP @ NAACL 2024**
- **Aryo Pradipta Gema**, Giwon Hong, Pasquale Minervini, Luke Daines, Beatrice Alex. (2024). Edinburgh clinical NLP at SemEval-2024 Task 2: Fine-tune your model unless you have access to GPT-4. **SemEval 2024**
- **Aryo Pradipta Gema**, Pasquale Minervini, Luke Daines, Tom Hope, Beatrice Alex. (2024). Parameter-efficient fine-tuning of llama for the clinical domain. **ClinicalNLP @ NAACL 2024**
- Imane Guellil, Jinge Wu, **Aryo Pradipta Gema**, ..., Honghan Wu, Bruce Guthrie, Beatrice Alex. (2023). *Natural Language Processing for Detecting Adverse Drug Events: Systematic Review*. **NIHR Open Research**
- Noptovius Halimawan, Derwin Suhartono, **Aryo Pradipta Gema**, Rezki Yunanda. (2022). BERT and ULMFiT Ensemble for Personality Prediction from Indonesian Social Media Text. **ISITDI 2022**
- David William, Said Achmad, Derwin Suhartono, **Aryo Pradipta Gema** (2022). Leveraging BERT with Extractive Summarization for Depression Detection on Social Media. **ISITIA 2022**
- Fernandes Saputra, Un Greffin Namyu, Derwin Suhartono, **Aryo Pradipta Gema** (2021). Automatic Piano Sheet Music Transcription with Machine Learning. **Journal of Computer Science**
- Derwin Suhartono, **Aryo Pradipta Gema**, Suhendro Winton, Theodorus David, Mohamad Ivan Fanany, Aniati Murni Arymurthy (2020). Sequence-To-Sequence Learning For Motion-Aware Claim Generation. **IJComp 2020**
- Derwin Suhartono, **Aryo Pradipta Gema**, Suhendro Winton, Theodorus David, Mohamad Ivan Fanany, Aniati Murni Arymurthy (2020). Argument annotation and analysis using deep learning with attention mechanism in Bahasa Indonesia. **Journal of Big Data**
- Derwin Suhartono, **Aryo Pradipta Gema**, Suhendro Winton, Theodorus David, Mohamad Ivan Fanany, Aniati Murni Arymurthy (2019). Attention-based argumentation mining. **International Journal of Computational Vision and Robotics**
- Ricky Anderson, **Aryo Pradipta Gema**, Sani M Isa, others (2018). Facial attractiveness classification using deep learning. **INAPR 2018**
- Derwin Suhartono, **Aryo Pradipta Gema**, Suhendro Winton, Theodorus David, Mohamad Ivan Fanany, Aniati Murni Arymurthy (2017). Hierarchical attention network with XGBoost for recognizing insufficiently supported argument. **MIWAI 2017**
- **Aryo Pradipta Gema**, Suhendro Winton, Theodorus David, Derwin Suhartono, Muhsin Shodiq, Wikaria Gazali (2017). It takes two to tango: modification of siamese long short term memory network with attention mechanism in recognizing argumentative relations in persuasive essays. **Procedia computer science**

Preprint (in submission)

- Igor Shilov, Alex Cloud, **Aryo Pradipta Gema**, ..., Erik Jones, Cem Anil (2025). *Beyond Data Filtering: Knowledge Localization for Capability Removal in LLMs*. arXiv preprint arXiv:2512.05648
- Joshua Ong Jun Leang, Zheng Zhao, **Aryo Pradipta Gema**, Sohee Yang, ..., Shay B Cohen (2025). *PiCSAR: Probabilistic Confidence Selection And Ranking*. arXiv preprint arXiv:2508.21787
- Alexander Murphy, ..., **Aryo Pradipta Gema**, Pasquale Minervini (2025). *An Analysis of Decoding Methods for LLM-based Agents for Faithful Multi-Hop Question Answering*. arXiv preprint arXiv:2503.23415

- Giwon Hong, **Aryo Pradipta Gema**, ..., Clémentine Fourier, Pasquale Minervini (2024). *The Hallucinations Leaderboard—An Open Effort to Measure Hallucinations in Large Language Models*. arXiv preprint arXiv:2404.05904.
- Giorgos Lysandrou, ..., **Aryo Pradipta Gema**, Beatrice Alex, Elizabeth Fairley (2024). *A Comparative Study on Patient Language across Therapeutic Domains for Effective Patient Voice Classification in Online Health Discussions*. arXiv preprint arXiv:2407.16593.
- **Aryo Pradipta Gema**, Michał Kobiela, Achille Fraise, Ajitha Rajan, Diego Oyarzun, Javier Alfaro (2023). *Vaxformer: Antigenicity-controlled Transformer for Vaccine Design Against SARS-CoV-2*. arXiv preprint arXiv:2305.11194.

INVITED TALKS

- **Inverse Scaling in Test-Time Compute**, Mar 2026, EPFL
- **Inverse Scaling in Test-Time Compute**, Jan 2026, EleutherAI
- **Inverse Scaling in Test-Time Compute**, Sep 2025, AI Safety Hub Edinburgh
- **DeCoRe: Decoding by Contrasting Retrieval Heads to Mitigate Hallucination**, May 2025, King's College London
- **DeCoRe: Decoding by Contrasting Retrieval Heads to Mitigate Hallucination**, Feb 2025, Alan Turing Institute
- **PhD in LLM Era**, Jan 2025, Bina Nusantara University
- **DeCoRe: Decoding by Contrasting Retrieval Heads to Mitigate Hallucination**, Dec 2024, Centre for AI @AstraZeneca
- **PhD in Clinical NLP**, Nov 2024, Epigene Labs
- **Clinical NLP in the LLM era**, Jul 2024, Discovery Sciences @AstraZeneca
- **Clinical NLP in the LLM era**, Apr 2024, Clinical NLP Group @University of Edinburgh
- **Clinical NLP in the LLM era**, Apr 2024, CDT Biomedical AI @University of Edinburgh
- **Parameter-efficient Fine-tuning of LLMs for the Clinical Domain**, Dec 2023, Bina Nusantara University
- **Parameter-efficient Fine-tuning of LLMs for the Clinical Domain**, Nov 2023, Bina Nusantara University
- **Knowledge Graph Embeddings in the Biomedical Domain**, Nov 2023, University of Edinburgh
- **Parameter-efficient Fine-tuning of LLMs for the Clinical Domain**, Oct 2023, University of Edinburgh
- **Introduction to the Large Language Model**, Jul 2023, Sutton Trust Summer School
- **Knowledge Graph Embeddings in the Biomedical Domain**, May 2023, Bina Nusantara University
- **Knowledge-Augmented Clinical Language Models**, May 2023, Bina Nusantara University
- **ChatGPT: A Biomedical AI Perspective**, May 2023, University of Edinburgh
- **Knowledge-Augmented Clinical Language Models**, Mar 2023, Bina Nusantara University
- **Clinical Language Models**, Feb 2023, Bina Nusantara University
- **Building a Knowledge Base from Biomedical Literature**, Jul, 2022, Bina Nusantara University
- **Augmenting Intelligence to Drug Discovery**, Dec, 2021, Bina Nusantara University
- **Augmenting Intelligence to Drug Discovery**, Apr, 2021, Bina Nusantara University
- **Introduction to Artificial Intelligence**, Oct, 2020, Bina Nusantara University

PEER REVIEW

- ACL Rolling Review - October 2023, December 2023 (**Outstanding reviewer EACL 2024**), February 2024, April 2024, June 2024 (**Outstanding reviewer EMNLP 2024**), August 2024, October 2024, December 2024, February 2025, May 2025, July 2025
- BlackboxNLP @ EMNLP 2025 - September 2025
- Springer Nature Machine Learning - June 2025
- COLM - May 2025
- 8th Healthcare Text Analytics Conference - March 2025
- Conference in AI for Health 2025 - March 2025
- Reasoning and Planning for LLMs @ ICLR 2025 - February 2025
- ICLR 2025 - October 2024
- 7th Healthcare Text Analytics Conference - March 2024
- Conference in AI for Health 2024 - March 2024
- Clinical NLP at NAACL 2024 - February 2024

AWARDS & ACHIEVEMENTS

Academic

- **Binusian Award (2018):** awarded to the most achieving graduate.
- **Applied Research Grant Binus (2017):** awarded to research initiatives by Binus University.
- **Scholarship Mentor Program (2015):** awarded to students that successfully mentor their peers.
- **Student of Excellence Honour (2015):** awarded to students with international and national achievements.
- **Student of Excellence Honour (2014):** awarded to students with international and national achievements.

Non-academic

- **United Asian Debating Championship (2015):** Quarterfinalist in English as Foreign Language Category.
- **Asian Law Students' Association University of Indonesia (2015):** Octofinalist.
- **Asian Law Students' Association Universitas Padjadjaran (2015):** Semifinalist and 8th Best Speaker.
- **Indonesian Varsity English Debate University of Indonesia (2015):** Champion in Novice Category.
- **Founder's Trophy Debating Championship University of Indonesia (2014):** Best Novice Team.
- **Newbies Debating Championship University of Indonesia (2014):** 1st Runner Up and 3rd Best Speaker.

SKILLS

Languages: Indonesian – Native, Banjarese - Native, English – Bilingual, French – Elementary

Programming Languages: Python, C/C++, Go, JavaScript, SQL

Libraries: PyTorch, TensorFlow, Transformers, Accelerate, Scikit-Learn, NumPy, Pandas, Matplotlib, Seaborn

Technologies: Git, Slurm, Docker, Kubernetes, Flask, Streamlit, MySQL, PostgreSQL, MongoDB, Azure Cosmos DB

Other interests: Chess, Parliamentary Debate, Sports, Cooking